

Auction-Dynamics MBO on Surface Meshes

The threshold-dynamics replacement for Phase 1: provenance, derivation, what transfers from the literature, and what does not

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Abstract

Approach B (`src/partition/mbo.auction.py`) replaces the Phase 1 Γ -convergence relaxation by volume-constrained multiphase threshold dynamics: the state is a hard labelling of the mesh vertices, each step diffuses every cell's indicator by one backward-Euler step of the surface heat equation and reassigns every vertex by a balanced thresholding $\arg \max_k [y_{ik} + \psi_k]$ whose offsets ψ enforce equal discrete area. None of the ingredients is new: the scheme is Merriman–Bence–Osher's, its variational interpretation is Esedoğlu–Otto's, the balanced thresholding is Jacobs–Merkurjev–Esedoğlu's auction dynamics, and the implicit-Euler diffusion is the one the graph-MBO literature uses. This document first records, from the papers themselves, what each of those works did, at what cell counts, on what domains — because an earlier description of this approach overstated its novelty on three of its four clauses. It then derives what is specific to this implementation: that the backward-Euler step is an *exponential mixture* of heat kernels, from which its interface constant ($\frac{1}{2}$, against $1/\sqrt{\pi}$ for the Gaussian) and its per-step displacement ($\tau\kappa/2$, against $\tau\kappa$) follow in three lines each and are confirmed on the production mesh; that the balanced threshold is the dual of a transportation problem whose two-phase case is the Ruuth–Wetton shifted level; that the Esedoğlu–Otto dissipation theorem does *not* apply to the implemented pairing of a consistent diffusion with a lumped assignment and an inexact dual, and exactly which inequality survives; that a lumped–lumped pairing would restore the theorem (not implemented); and the two-sided time-step window, with the over-merge side quantified by the diffused indicator's value at a cell's centre, $1 - xK_1(x)$ at $x = R_{\text{cell}}/\sqrt{\tau}$. Measurements are cited from report 08; nothing here is a new measurement except the one-step displacement check.

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1 Overview

Provenance of every computed number in this document

Status	derivation, complete; twice adversarially reviewed (2026-09-17), all findings applied
Script	<code>docs/math/10-mbo-auction-dynamics/check_numerics.py</code> → committed <code>numerics.yaml</code> beside this file. Every number below that was produced by running code is quoted from that file , with its key named in the text; nothing is quoted from an interactive session or from a reviewer’s scratch script. Run from the repo root (≈ 5 min; <code>--quick</code> skips the two 100×96 dense eigendecompositions).
Determinism	jittered disc centres (§5.6): <code>numpy.random.default_rng(0)</code> , 24 samples per row, one shared generator consumed row by row in the table’s order, offsets $\theta \sim U(0, 2\pi/348)$, $\varphi \sim U(0, 2\pi/328)$. Labelling pair (§7.2): the geodesic balanced init at the project seed 84172851, then one full-budget balanced step.
Meshes	torus $R = 1.0$, $r = 0.6$ throughout (<code>TorusMeshProvider</code>); 40×36 , 60×52 , 100×96 (the production base), and the 348×328 deliverable mesh ($V = 114,144$); τ from <code>tau_diagnostics</code> with <code>MBOConfig()</code> defaults.
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1; macOS 15.3.1 arm64 (Mac mini). Eigenvalues may differ in the last printed digit across BLAS builds; nothing quoted depends on that digit.
Cited, not computed	the run statistics of §7.2 and §7.3 (4 of 96 energy increases; the slack range) are from <code>docs/experiments/08-mbo-auction-dynamics/data.yaml</code> ; the 68.16%/1.97% of §9 from <code>docs/plans/PHASE1_BC.REPLACEMENT.PLAN.md</code> ; the freeze constants’ $c = 2/c = 4$ pair from <code>testing/test_mbo_auction.py</code> ; every statement about a paper from the copy indexed in <code>docs/papers/README.md</code> .

Phase 1 of this project, as specified by Bogosel and Oudet [5] and derived in document 06 (`docs/math/06-phase1-energy-discretization/`), minimises a Modica–Mortola functional over N continuous densities under an equal-mass constraint, and reads the partition off as the winner-take-all territory of those densities. Document 09 (`docs/math/09-balanced-readout/`) derives why that readout can diverge from what was constrained — at high N it manufactures cells whose territory is far from the equal-area target, and cells whose territory is disconnected — and the extraction-time correction that repairs it.

Approach B removes the category of problem rather than repairing it. Its state is a labelling $\omega : \{1, \dots, V\} \rightarrow \{1, \dots, N\}$ of the mesh vertices, never a continuous field, so there is no readout to diverge. Each step at a fixed mesh level is

1. **diffuse**: for every cell k , solve $(M + \tau K) y_k = M \chi_k$, where χ_k is the indicator of cell k ’s territory;
2. **balanced threshold**: $\omega'(i) = \arg \max_k [y_{ik} + \psi_k]$, with the N offsets ψ chosen so that every cell’s discrete area equals the target $\bar{A}$.

This is repeated until the labels stop changing, the labels are carried to the next finer mesh by nearest-vertex interpolation, and the process repeats. Phase 2 then refines the resulting contour exactly as it would a Phase 1 output (`src/optimization/perimeter_optimizer.py`); nothing downstream changed.

The measurements — perimeter against the validated PGD deliverables, the three validity gates, attribution to the dynamics rather than the initialisation, seed spread, and the $N = 400$ – 1000 runs — are in `docs/experiments/08-mbo-auction-dynamics/` and are not repeated here. This document is the derivation and the provenance. Following the scope policy of `docs/math/AUTHORING_GUIDE.md`, it derives only what `src/partition/mbo_auction.py` computes; where a cleaner variant exists on paper (§7.4) it is stated and marked as not implemented.

2 Provenance: what was done before, and what is claimed here

Every reference in this section was read from the copy held in docs/papers/ (title page, abstract, and the sections cited), not from a secondary description. The two exceptions are marked.

2.1 The lineage

Threshold dynamics. Merriman, Bence and Osher introduced the scheme in a 1992 UCLA report [20]: diffuse the indicator of a set for a short time, threshold at $\frac{1}{2}$, repeat; the boundary moves by mean curvature. Their 1994 paper [21] extends it to several phases by diffusing every indicator and assigning each point to the largest — the $\arg \max$ step above. Convergence of the two-phase scheme to mean-curvature flow was proved by Evans [14]; the multiphase case, in the sense of a BV weak formulation and conditionally on energy convergence, by Laux and Otto [18]. Ruuth [23] extended the multiphase scheme to prescribed junction angles.

The variational interpretation. Esedoğlu and Otto [13] showed that one MBO step is the minimiser of the *linearisation* of a “heat content” energy E_τ at the current configuration, that this energy is concave on the convex hull of partitions, so the step never increases it (their Lemma 5.2, for any kernel $G \geq 0$ with $\hat{G} \geq 0$, §5.2), and that E_τ Γ -converges to the total interfacial energy as $\tau \rightarrow 0$ (their Appendix A, for smooth radial kernels). Esedoğlu and Jacobs [12] widen the Γ -convergence to kernels that are merely nonnegative, symmetric, integrable with a finite first moment, and positive near the origin (their Theorem 5.1), which is the class the present kernel belongs to (§5.4). Alberti and Bellettini’s nonlocal phase-transition model is the underlying theory; the held Part I [2] is the optimal-profile paper, and the Γ -limit itself is in its companion (European J. Appl. Math. 9, 1998), which is not held.

Volume constraints. Ruuth and Wetton [24] enforce a prescribed volume in the two-phase scheme by shifting the threshold level away from $\frac{1}{2}$ until the volume is met. (*Cited second-hand: the paper is paywalled; its content is taken from the descriptions in [17, §2] and [19], which proves its convergence, and the bibliographic details from the publisher’s listing.*) Jacobs, Merkurjev and Esedoğlu [17] generalise this to N phases: minimising the linearisation of E_τ under volume constraints is a linear assignment problem, its Lagrange multipliers shift the thresholds (their eq. (11), which is exactly the $\arg \max_k [y_{ik} + \psi_k]$ of this implementation), and they solve it with Bertsekas’ auction algorithm [4]; hence “auction dynamics”. Laux and Swartz [19] prove convergence of the volume-preserving thresholding scheme.

Graphs, and the implicit-Euler diffusion. Garcia-Cardona et al. [15] run multiclass MBO on a weighted graph and replace the Gaussian convolution by an *implicit-Euler* step of the graph heat equation, $U^{n+1/2} = B^{-1}[U^n - \delta t \mu(U^n - \hat{U})]$ with $B = I + \delta t L_s$ (their eqs. (27)–(28); eq. (29) is B in a truncated eigenbasis). Jacobs et al. [17, §5] run *volume-constrained* auction dynamics on an unstructured k -nearest-neighbour graph with unit-volume nodes — but with the symmetrised similarity matrix $W^\top W$ used *directly* as the kernel, no Laplacian and no implicit step. Van Gennip, Guillen, Osting and Bertozzi [25] prove the graph scheme is *pinned* for τ below a spectral bound and *trivial* above another (their Theorems 4.2–4.4) — the rigorous form of the two-sided window of §8.

Surfaces. Merriman and Ruuth [22] move curves with junctions on a curved surface by extending each indicator off the surface with the closest-point map, diffusing in $\mathbb{R}^3$ on a Cartesian grid, and thresholding on the surface; their §8.3 evolves five regions on an embedded torus and, in their Fig. 11, the 174 countries plus ocean of a world map on the sphere — many-region multiphase threshold dynamics on a curved closed surface, unconstrained. Wang and Osting [27] compute *Dirichlet* partitions (minimal sum of first Laplace–Beltrami eigenvalues, not perimeter)

of flat tori and of the sphere by a diffusion-generated method, via FFT and the spherical harmonic transform, for $k \leq 20$. The surface finite-element setting used here — P1 mass and stiffness matrices on a triangulation of an embedded surface — is Dziuk’s [8, 9].

Partitions with many cells. Elsey, Esedoğlu and Smereka simulate grain growth by diffusion-generated motion with 166,927 grains in two dimensions [10] and 133,110 in three [11], on uniform grids, without volume constraints. Wang et al. [26] apply iterative convolution–thresholding to piecewise-constant Mumford–Shah segmentation with a handful of phases; Hu, Liu and Wang [16] use auction dynamics as the inner solver of the *outer* (Pólya) problem on 256^2 and 128^3 grids with $n \leq 9$ cells. Bogosel and Oudet [6] initialise the same outer problem from capacity-constrained Voronoi diagrams with more than 100 equal-area cells in the plane.

Work	Domain, discretisation	Cells	Equal-area	Relation to this implementation
[17] §4.2	flat 2-torus $[0, 1]^2$, grid, FFT	17, 64 ; 160	yes (160: pre-served)	same step, same problem class, exact auction
[17] §4.2	flat 3-torus	8; 32	yes (32: pre-served)	Weaire–Phelan found with temperature
[22] §8.3	embedded torus and sphere, closest point, grid	5 (torus); 175 (sphere)	no	multiphase MBO on curved closed surfaces
[27]	flat tori and S^2 , FFT / SHT	≤ 20	no	other energy; curved closed surface
[16]	box, 256^2 / 128^3 grid	≤ 9	yes	auction dynamics as inner solver
[15]	weighted graph, eigenbasis	~ 10 classes	no	implicit-Euler diffusion step
[17] §5	unstructured k NN graph, kernel $W^\top W$	~ 10 classes	bounds	constrained + unstructured, no surface
[10, 11]	grids	10^5	no	scale, unconstrained
[6]	plane	> 100 (init only)	yes	capacity-constrained Voronoi as init
[5]	meshed surfaces, Γ -convergence	≤ 11 torus, ≤ 32 sphere	yes	the Phase 1 incumbent; no thresholding

Table 1: Cell counts and settings, read from the cited sections. “Preserved” means each Voronoi cell keeps its *initial* area or volume (160 cells in 2D, 32 grains in 3D), not that they are equal; “init only” means the count appears in an initialisation figure, not a minimisation run.

2.2 The corrected statement of what is new

An earlier description of approach B (the module docstring, and §2 of report 08 as first written) said that every published implementation of this machinery runs “on a uniform Cartesian grid via FFT, on a flat domain, at small N , for Pólya’s outer problem”. Table 1 falsifies three of those four clauses: [17] §4.2 solves the *inner* problem at $N = 64$ and 160 (two clauses), and [22] runs the multiphase scheme on a curved torus and on the sphere (the “flat” clause; JME’s torus is flat). Only “uniform grid via FFT” survives, and only for the Euclidean implementations; the graph line never used an FFT.

What the held literature does *not* contain, stated as “we are not aware of” and not as “first”:

1. the scheme on a **curved, embedded surface discretised by an unstructured triangle mesh**, with the diffusion done by a surface-FEM backward-Euler step — the graph line has the operator form, the closest-point line has the surface, neither has both, and neither works with lumped and consistent mass on a triangulation;
2. **equal-area constraints at $N = 400$ – 1000** : in the held papers, equal-area minimisation

reaches 64 cells [17, §4.2]; the 160 and 32 there are area-preserving, and the “more than 100” of [6] is an initialisation figure;

3. the assignment solved by an **inexact** transportation dual (§6) rather than an exact auction, together with the inequality that survives inexactness (§7.3);
4. the **multilevel mesh ladder** with label interpolation and a per-level, mesh-derived time step (§10, §8);
5. coupling to an exact contour refinement (Phase 2) and the gate instruments of report 07.

The defensible framing is therefore *combination, surface, and scale*, not *method*. The method is [17]’s; the diffusion operator is the graph-MBO one transplanted to surface finite elements; “large N ” by itself is not new in three separate lines of work. And the held papers come closer than a one-line summary suggests: [17, §5] already has the volume constraint on an *unstructured* domain (a similarity graph), [22, Fig. 11] has 175 regions on a curved closed surface, and [27] has a curved closed surface with exact spectral diffusion at 20 cells. What none of them has is all of it at once on a triangulated surface with equal-area constraints in the hundreds.

3 Notation and Setup

The surface S is closed and triangulated with vertices $\mathbf{V} = \{x_1, \dots, x_V\}$ and faces $\mathbf{F}$. M and K are the P1 mass and stiffness matrices of docs/math/06-phase1-energy-discretization/eq. (MK-def), assembled by `TriMesh.compute_matrices`; K is the cotangent Laplacian and $K\mathbf{1} = 0$. The lumped mass is $v_i = \sum_j M_{ij} > 0$, $D = \text{diag}(v)$, $|S| = \sum_i v_i$, and the equal-area target is $\bar{A} = |S|/N$. The mesh scales are the mean and maximum edge length h_{mean} , h_{max} over all triangle sides, and the cell scale is the radius of the equal-area disc,

$$R_{\text{cell}} = \sqrt{\bar{A}/\pi}. \quad (1)$$

A labelling ω has indicator columns $\chi_k \in \{0, 1\}^V$, $\chi_{ik} = [\omega(i) = k]$, so $\sum_k \chi_k = \mathbf{1}$, and territories $\Omega_k = \{i : \omega(i) = k\}$ with discrete areas $T_k = \sum_{i \in \Omega_k} v_i = v^\top \chi_k$. The diffused indicators are the columns y_k of $y \in \mathbb{R}^{V \times N}$. In the continuum, G_t is the heat kernel of S at time t (on $\mathbb{R}^2$, $G_t(x) = (4\pi t)^{-1} e^{-|x|^2/4t}$), and K_τ^{res} the kernel of $(I - \tau \Delta)^{-1}$, defined in §5.4.

Code: the objects

```
src/partition/mbo_auction.py:      sparse_one_hot(labels, n_cells)      builds       $\chi$ ;  
edge_lengths(mesh) gives  $h_{\text{mean}}, h_{\text{max}}$ ; cell_radius(total_area, n_cells) is (1); mesh.v is  $v$ ,  
mesh.M, mesh.K are  $M, K$ .
```

4 Threshold dynamics in the continuum

4.1 The MBO scheme

For a set $\Sigma \subset S$ and a time step τ , one step of [20] is

$$\Sigma' = \{x : (G_\tau * \chi_\Sigma)(x) \geq \tfrac{1}{2}\}, \quad (2)$$

and for a partition $(\Omega_1, \dots, \Omega_N)$ one step of [21] is

$$\omega'(x) = \arg \max_k (G_\tau * \chi_{\Omega_k})(x). \quad (3)$$

Since $\sum_k G_\tau * \chi_{\Omega_k} = 1$, (3) reduces to (2) when $N = 2$. Away from junctions, each interface moves by mean curvature over time τ per step, to leading order in τ [14, 13]; §5.5 derives the constant for the kernel actually used here.

4.2 The heat-content energy and the linearisation principle

Following [13, §4], define for a partition

$$E_\tau(\omega) = \frac{1}{\sqrt{\tau}} \sum_{k=1}^N \int_S \chi_{\Omega_k} (1 - G_\tau * \chi_{\Omega_k}) dA = \frac{1}{\sqrt{\tau}} \sum_{k \neq j} \int_S \chi_{\Omega_k} G_\tau * \chi_{\Omega_j} dA, \quad (4)$$

the second form by $\sum_j G_\tau * \chi_{\Omega_j} = 1$. Each term measures the heat that escapes from cell k into the others in time τ — “heat content”. Note the sum is over *ordered* pairs, so every interface is counted from both sides; this is eq. (4.2) of [13] with all surface tensions equal to one and the same $\tau^{-1/2}$ normalisation.

Relaxed to $u : S \rightarrow \Delta_N$ (the probability simplex), the energy is

$$E_\tau(u) = \frac{1}{\sqrt{\tau}} \left[|S| - \sum_k \int_S u_k G_\tau * u_k dA \right], \quad (5)$$

which is **concave**: $u_k \mapsto \int u_k G_\tau * u_k$ is a quadratic form with a positive semi-definite kernel ($\widehat{G_\tau} = e^{-\tau|\xi|^2} \geq 0$), so its negative is concave. Its linearisation at u is, up to a constant,

$$L_u(u') = -\frac{2}{\sqrt{\tau}} \sum_k \int_S u'_k G_\tau * u_k dA, \quad (6)$$

and minimising L_u over simplex-valued u' is a pointwise problem whose solution puts all of $u'(x)$ on the largest $(G_\tau * u_k)(x)$ — i.e. exactly the step (3). This is the observation of [13] that makes threshold dynamics a minimising-movements scheme, and it carries the dissipation property:

Proposition 4.1 (Dissipation; [13, Lemma 5.2]). *Let F be concave and differentiable on a closed convex set $\mathcal{K}$, and let $u' \in \arg \min_{\mathcal{K}} L_u$ where L_u is the linearisation of F at $u \in \mathcal{K}$. Then $F(u') \leq F(u)$.*

Proof. Concavity gives $F(u') \leq L_u(u')$; optimality of u' and feasibility of u give $L_u(u') \leq L_u(u) = F(u)$. $\square$

The proof uses only that u itself lies in $\mathcal{K}$. So the conclusion survives *any* further convex constraint on u' that u already satisfies — in particular the equal-area constraint of §6, provided the constrained linearisation is minimised *exactly*. That is the route by which [17] obtain dissipation for auction dynamics, and §7 examines which of its hypotheses the implementation violates.

4.3 Γ -convergence and the interface constant

For a kernel $K_\tau(x) = \tau^{-d/2} K(x/\sqrt{\tau})$ with $K \geq 0$, radial, $\int K = 1$ and a finite first moment, the heat content of a smooth set satisfies, per unit interface length and to leading order in τ ,

$$\frac{1}{\sqrt{\tau}} \int_{\Sigma^c} K_\tau * \chi_\Sigma dA \longrightarrow c_K \text{Per}(\Sigma), \quad c_K = \frac{1}{2} \int_{\mathbb{R}^d} |z_1| K(z) dz = \int_0^\infty r k(r) dr, \quad (7)$$

where k is the one-dimensional marginal of K . The second form is the flat calculation: for an interface $\{x_1 = 0\}$, the pairs (s, t) with $s < 0 < t$ and $t - s = r$ have measure r , so the escaped heat per unit length is $\int_0^\infty r k_\tau(r) dr = \sqrt{\tau} \int_0^\infty r k(r) dr$. Summed over cells in (4), every interface is counted from both sides, so

$$E_\tau(\omega) \longrightarrow c_K \sum_k \text{Per}(\Omega_k). \quad (8)$$

For the Gaussian, k is $N(0, 2)$ and $c_G = \frac{1}{2}\sqrt{4/\pi} = 1/\sqrt{\pi}$. This is the constant $c_0 = (|B^{d-1}|/|S^{d-1}|) \int |z| G(z) dz$ of [13, eq. (A.2)]: in $d = 2$ it is $\pi^{-1}\sqrt{\pi} = 1/\sqrt{\pi}$, and the same value follows from (7) because $\int |z_1| K = (|B^{d-1}|/|S^{d-1}|) \cdot 2 \int |z| K$ for a radial kernel. The rigorous statement — Γ -convergence of (5) in L^1 to $c_K \sum_k \text{Per}(\Omega_k)$, with compactness — is [13, Proposition A.1] for kernels that are smooth, nonnegative, radial, radially non-increasing, of unit mass with a finite first moment, and satisfy $|\nabla G(z)| \lesssim G(z/2)$; and [12, Theorem 5.1] for kernels that are merely nonnegative, symmetric, in L^1 with $|z|K \in L^1$, and positive in a neighbourhood of the origin. The kernel used here fails the first set (it is singular at the origin, §5.4) and satisfies the second. Both theorems are proved on a periodic box in $\mathbb{R}^d$; neither covers a curved surface or a finite-element discretisation. That transfer is assumed here, not proved.

5 The surface finite-element diffusion step

5.1 One backward-Euler step

The semi-discrete P1 heat equation on the surface is $M\dot{y} + Ky = 0$ [9]. One backward-Euler step of length τ from $y(0) = \chi_k$ is

$$(M + \tau K) y_k = M \chi_k, \quad \text{i.e.} \quad y_k = A_\tau \chi_k, \quad A_\tau = (M + \tau K)^{-1} M. \quad (9)$$

This is the only linear solve in the method. $M + \tau K$ is symmetric positive definite, is factorised once per (level, τ) pass — in the step loop, again for the probe baseline and each probe factor, and for the final full-budget assignment — and the N right-hand sides are back-substituted.

Code: the diffusion step

```
src/partition/mbo-auction.py: make_diffusion_solver(mesh, tau) returns factorized((M
+ tau*K).tocsc()) (SuperLU); diffuse_indicators(solve, mesh, labels, n_cells) forms
Mχ sparsely and solves column by column, accumulating  $\sum_k (M\chi_k)^\top y_k$  for §11.
```

5.2 The step is an exponential mixture of heat semigroups

Everything that is specific to this kernel follows from one identity. Let $L = M^{-1}K$; it is diagonalisable with real eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots$ by the generalised eigenproblem $K\phi = \lambda M\phi$, and e^{-tL} is the semi-discrete heat semigroup.

Proposition 5.1 (Mixture representation).

$$A_\tau = (I + \tau L)^{-1} = \int_0^\infty \frac{1}{\tau} e^{-t/\tau} e^{-tL} dt, \quad (10)$$

i.e. one backward-Euler step is the heat semigroup averaged over an exponentially distributed time $t \sim \text{Exp}(\text{mean } \tau)$. The same identity holds in the continuum, $(I - \tau \Delta)^{-1} = \int_0^\infty \tau^{-1} e^{-t/\tau} e^{t\Delta} dt$.

Proof. On each eigenvector, $\int_0^\infty \tau^{-1} e^{-t/\tau} e^{-t\lambda} dt = \tau^{-1}(\tau^{-1} + \lambda)^{-1} = (1 + \tau\lambda)^{-1}$; the eigenvectors span $\mathbb{R}^V$. $\square$

5.3 Structural properties

Proposition 5.2. (i) **Partition of unity.** $\sum_k y_k = \mathbf{1}$ exactly, for every labelling.

(ii) **Symmetry.** $MA_\tau = M(M + \tau K)^{-1}M$ is symmetric positive definite, so $\chi \mapsto -\chi^\top MA_\tau \chi$ is concave.

(iii) **Positivity is not guaranteed on the mesh.** In the continuum A_τ has a positive kernel (§5.4). On a triangulation, K has positive off-diagonal entries across obtuse angles and M is not diagonal, so e^{-tL} need not be entrywise nonnegative (on a 40×36 torus mesh its minimum entry is -1.4×10^{-2} at $t = 0.1h^2$), and no argument gives nonnegativity of the mixture A_τ either. In practice it holds: on every torus mesh tested at the production τ the minimum entry of A_τ is positive ($+2.7 \times 10^{-5}$ at 40×36 , $N = 10$; $+8.7 \times 10^{-9}$ at 60×52 , $N = 100$; $+3 \times 10^{-10}$ and 0 to twelve digits at 100×96 , $N = 100$ and 400), so the caveat is inactive on the ladder. (numerics.yaml: positivity_and_pairing.exp_m_check, pairing_cases[*].min_entry_A_tau.)

Proof. (i) $K\mathbf{1} = 0$ gives $(M + \tau K)\mathbf{1} = M\mathbf{1}$, so $A_\tau\mathbf{1} = \mathbf{1}$, and $\sum_k \chi_k = \mathbf{1}$. (ii) $(M + \tau K)^{-1}$ is SPD and M is symmetric, so $M(M + \tau K)^{-1}M$ is SPD. (iii) is the standard failure of the discrete maximum principle for the cotangent Laplacian on non-Delaunay meshes, and the numbers are direct computation. Nothing in the method relies on positivity: the assignment is invariant under adding a common constant to a vertex's scores, and the balanced threshold does not compare against $\frac{1}{2}$. $\square$

5.4 The continuum kernel and its interface constant

By (10) the kernel of $(I - \tau\Delta)^{-1}$ on $\mathbb{R}^2$ is

$$K_\tau^{\text{res}}(r) = \int_0^\infty \frac{1}{\tau} e^{-t/\tau} \frac{e^{-r^2/4t}}{4\pi t} dt = \frac{1}{2\pi\tau} K_0\left(\frac{r}{\sqrt{\tau}}\right), \quad (11)$$

with K_0 the modified Bessel function. From the mixture, without evaluating (11): K_τ^{res} is nonnegative, radial, of unit mass, has all moments $\int |z|^p K_\tau^{\text{res}} = \mathbb{E}_t[\int |z|^p G_t] \propto \mathbb{E}[t^{p/2}] < \infty$, and is positive (indeed logarithmically singular) at the origin. It therefore lies in the class of [12, Theorem 5.1]; its Fourier symbol $(1 + \tau|\xi|^2)^{-1}$ is positive, so it also satisfies the hypothesis of Proposition 4.1's continuum version [13, §5.2]. Its tails are exponential rather than Gaussian: the one-dimensional marginal is

$$k_\tau(s) = \frac{1}{2\sqrt{\tau}} e^{-|s|/\sqrt{\tau}}, \quad (12)$$

because restricting the symbol to the ξ_1 -axis gives $(1 + \tau\xi_1^2)^{-1}$, the transform of (12).

Proposition 5.3 (Interface constant of the backward-Euler kernel). $c_{K_\tau^{\text{res}}} = \frac{1}{2}$. Hence, for the kernel actually used,

$$E_\tau(\omega) \longrightarrow \frac{1}{2} \sum_k \text{Per}(\Omega_k) = \text{total interface length, each interface counted once.} \quad (13)$$

Proof. Either from (12): $\int_0^\infty r k_1(r) dr = \frac{1}{2} \int_0^\infty r e^{-r} dr = \frac{1}{2}$. Or from the mixture: the Gaussian at time t escapes $\sqrt{t/\pi}$ per unit length, and $\mathbb{E}[\sqrt{t}] = \sqrt{\tau} \Gamma(\frac{3}{2}) = \frac{1}{2} \sqrt{\pi\tau}$, so $\tau^{-1/2} \mathbb{E}[\sqrt{t/\pi}] = \frac{1}{2}$. $\square$

Remark 5.1 (The factor of two in the perimeter convention). Every perimeter this project reports — `final_perimeter`, `arm_report.yaml`, the deliverable tables — is $\sum_k \text{Per}(\Omega_k)$, which counts each interface twice. By (13), E_τ approximates *half* of that quantity, i.e. the same thing `label_boundary_length` measures (§11). With the Gaussian it would approximate $(2/\sqrt{\pi}) \approx 1.128$ times the interface length instead.

5.5 The per-step displacement

Proposition 5.4 (Displacement per step). *Let Σ have a smooth boundary with (geodesic) curvature κ at a point, positive when Σ is locally convex. One step (2) with the Gaussian G_τ moves the boundary inward by $\tau\kappa$; with the backward-Euler kernel K_τ^{res} it moves it by*

$$\delta = \frac{\tau\kappa}{2}, \quad (14)$$

both to leading order in τ . One backward-Euler MBO step is therefore mean-curvature flow over time $\tau/2$, not τ .

Proof. At signed distance d outside Σ , the Taylor expansion of the diffused indicator is [13, §5]

$$(G_t * \chi_\Sigma)(d) = \frac{1}{2} - \frac{d + t\kappa}{\sqrt{4\pi t}} + O(d^2, t), \quad (15)$$

so the Gaussian threshold set is $d = -t\kappa$. Averaging (15) over $t \sim \text{Exp}(\tau)$ and setting it to $\frac{1}{2}$,

$$d \mathbb{E}[t^{-1/2}] + \kappa \mathbb{E}[t^{1/2}] = 0 \implies d = -\kappa \frac{\mathbb{E}[t^{1/2}]}{\mathbb{E}[t^{-1/2}]} = -\kappa \frac{\sqrt{\tau} \Gamma(3/2)}{\tau^{-1/2} \Gamma(1/2)} = -\frac{\kappa\tau}{2}. \quad \square$$

Remark 5.2 (Second moments do not determine the speed). Both kernels have second moment 2τ per coordinate (expand $(1 + \tau|\xi|^2)^{-1}$ and $e^{-\tau|\xi|^2}$ to first order), so one might expect the same speed. The displacement is instead $\delta = \frac{\kappa}{2} \rho_K$ with $\rho_K = \int y_2^2 K(0, y_2) dy_2 / \int K(0, y_2) dy_2$ the ratio of the transverse second moment to the marginal density *on the interface*; ρ_K is 2τ for the Gaussian and τ for K_τ^{res} (a direct computation with (11), using $\int_0^\infty s^2 K_0(s) ds = \int_0^\infty K_0(s) ds = \pi/2$). The mixture proof avoids the Bessel integrals. Every identity in §5.2–§5.5 — the mixture against the closed-form kernel, both interface constants, the displacement by both routes, the Bessel integrals, the second moments, and the centre values of §8.3 — is evaluated numerically in `numerics.yaml: constants`.

5.6 Numerical check on the production mesh

Proposition 5.4 is for the continuum kernel and to leading order in τ ; the implementation uses (9) on a triangulation. For a disc the continuum problem is exactly solvable, which gives a reference that is not leading-order. With $x = R/\sqrt{\tau}$ and $u = r/\sqrt{\tau}$, the radial solution of $(1 - \tau\Delta)y = \chi_{B_R}$ is

$$y(u) = \begin{cases} 1 - xK_1(x)I_0(u), & u \leq x, \\ xI_1(x)K_0(u), & u \geq x, \end{cases} \quad (16)$$

(continuous with continuous derivative at $u = x$ by the Wronskian $I_0K_1 + I_1K_0 = 1/x$; the centre value is (24)), so the $\frac{1}{2}$ -threshold radius u^* and the exact area loss $\pi\tau(x^2 - u^{*2})$ follow by a one-dimensional root find. At the x of the table below it is $1.09\text{--}1.16 \times \pi\tau$: the leading-order $\pi\tau$ is low by 9–16% at these x , because the resolvent’s exponential tails make its $O(1/x)$ correction about four times the Gaussian’s (whose exact loss is $1.02\text{--}1.04 \times 2\pi\tau$ at the same x).

On the 348×328 torus mesh of the deliverables ($V = 114,144$, $h_{\text{mean}} = 0.0171$, $h_{\text{max}} = 0.0311$), take the set of vertices within *ambient* distance R_d of a point on the outer equator — not a geodesic disc; its lumped area differs from πR_d^2 by -1.4 to $+0.8\%$ — apply one step of (9) and a $\frac{1}{2}$ -threshold, and measure the lumped area change. The result depends on where the centre sits relative to the mesh: the boundary ring of vertices flips in groups, so a single placement carries granularity noise. The table gives the vertex-centred placement $(1.6, 0, 0)$ and the mean over 24 centres drawn uniformly within one grid cell around it, as ratios to the exact continuum loss (16) (all values: `numerics.yaml: disc_check`; seed and draw order in the provenance block):

R_d	τ	$\sqrt{\tau}/h_{\text{mean}}$	x	exact loss	vertex-centred	jittered mean $\pm$ std	vs. Gaussian exact
0.20	0.004	3.69	3.16	−0.01454	0.90	0.97 ± 0.06	0.54
0.25	0.004	3.69	3.95	−0.01392	0.89	0.96 ± 0.08	0.52
0.30	0.006	4.52	3.87	−0.02095	0.84	0.93 ± 0.04	0.50
0.20	0.002	2.61	4.47	−0.00684	0.77	0.91 ± 0.09	0.48

Averaged over placement the mesh reaches 0.91–0.97 of the exact continuum loss and 0.48–0.54 of the exact Gaussian value; the vertex-centred placement is the low tail of that distribution (an earlier version of this section reported it alone and read a 10–23% deficit into it). Of the remaining 3–9% shortfall, surface curvature accounts for 1.4–3.1% (the Gaussian curvature at the outer equator is $1/(r(R+r)) = 1.042$, which lowers the boundary’s geodesic curvature below $1/R_d$ by $KR_d^2/3$, i.e. 3.1% at $R_d = 0.3$); the rest is finite-element and threshold-granularity error at a per-step displacement of only 0.25–0.63 h_{mean} (implied by the measured vertex-centred loss, `displacement_over_h_mean`). What the check establishes is that the FEM step moves an interface at *half* the Gaussian rate to within the granularity of the test, as Proposition 5.4 says. This is a one-off check of the derivation, not a measured study.

6 The balanced threshold

6.1 The constrained linearisation is a transportation problem

Minimising the linearisation (6) under the equal-area constraint, discretised with a weight matrix W for the inner product, is

$$\max_{\chi'} \sum_k \langle \chi'_k, W y_k \rangle \quad \text{s.t.} \quad \sum_k \chi'_{ik} = 1 \quad \forall i, \quad \sum_i v_i \chi'_{ik} = \bar{A} \quad \forall k, \quad \chi'_{ik} \in \{0, 1\}. \quad (17)$$

The implementation takes $W = D$, so the objective is $\sum_i v_i y_{i, \omega'(i)}$ and (17) is the transportation problem (LP) of `docs/math/09-balanced-readout/ §6.2` with score matrix $s_{ik} = y_{ik}$: supplies v_i at the vertices, demands $\bar{A}$ at the cells. With unit vertex masses it would be the assignment problem of [17]; with the real lumped masses it is a transportation problem [1], which is why exact balance is generically unattainable (doc 09, §6.5) and why a granularity bar exists at all.

6.2 Dual, shifted argmax, and the two-phase antecedent

Doc 09 derives the dual of the LP relaxation and shows (its Proposition 6.1) that complementary slackness forces the assignment onto

$$\omega'(i) = \arg \max_k (y_{ik} + \psi_k), \quad (18)$$

in the code’s sign convention, where raising ψ_k grows cell k , and that the dual subgradient is $T_k - \bar{A}$: grow the cells that are too small. The ψ_k are the Lagrange multipliers λ^* of [17, eq. (11)] and the *prices* of the auction [4]. For $N = 2$, using $y_{i1} + y_{i2} = 1$,

$$y_{i1} + \psi_1 \geq y_{i2} + \psi_2 \iff y_{i1} \geq \frac{1}{2} + \frac{\psi_2 - \psi_1}{2}, \quad (19)$$

which is the threshold level shifted away from $\frac{1}{2}$ until the volume is met — the scheme of [24]. (18) is its N -phase form.

6.3 The inexact solver as implemented

`solve_dual_offsets` is a subgradient method on the piecewise-linear dual with a decaying step — descent on $D(\psi)$ in doc 09’s convention, ascent on the concave form in the code’s — returning the *best* iterate visited (doc 09, §6.2–6.3). Three things are specific to its use here:

1. **Scale.** The step schedule was calibrated on log-density scores. Diffused indicators live on $[0, 1]$, so with `normalize_scores=True` the scores are divided by the median gap between each vertex's top two scores (the margin an offset must cover to flip a vertex), the schedule `normalized_eta0 = 2.0`, `dual_decay = 0.02`, `normalized_dual_iters = 2000` is run on the normalised scores, and ψ is rescaled to the caller's units. Correctness of (18) is scale-free; only convergence is not.
2. **Quality bar.** A solve is accepted at $\max_k |T_k - \bar{A}|/\bar{A} \leq \text{bar}$ with $\text{bar} = \max(2\gamma, \text{A-parity})$, where $\gamma = \max_i v_i/\bar{A}$ is the one-vertex granularity and A-parity is approach A's own stall level at equal budget (report 07). A-parity is a two-entry lookup, for $(V, N) = (47, 488, 300)$ and $(114, 144, 300)$ only; for every other configuration — all $N = 100, 400\text{--}1000$ runs and every non-torus surface — the bar is exactly 2γ . It is a solver-failure detector, not a validity test: equal area is what the step optimises, so hitting it says the dual converged.
3. **Early stopping is production-only.** In-loop solves stop at the bar; every level ends with a full-budget re-assignment, and in-loop area figures are tagged `early_stop_capped` so they are never quoted as assignment quality (report 07, artefact 5).

Code: the balanced threshold

```
src/partition/mbo-auction.py:          balanced_assign(scores, lumped_mass, target,
bar, early_stop) → src/partition/balanced-readout.py: solve_dual_offsets with
BalancedReadoutConfig(normalize_scores=True, dual_early_stop_rel=bar or None);
assignment_margin_scale is the median top-two gap; src/partition/arm_harness.py:
assignment_quality_bar, vertex_granularity.
```

7 Energy dissipation: what transfers and what does not

7.1 The theorem, for the consistent pairing with an exact assignment

Discretise (5) with the consistent mass matrix:

$$E_\tau^M(\chi) = \frac{1}{\sqrt{\tau}} \left[|S| - \sum_k \chi_k^\top M A_\tau \chi_k \right] = \frac{1}{\sqrt{\tau}} \left[|S| - \sum_k \langle \chi_k, M y_k \rangle \right]. \quad (20)$$

By Proposition 5.2(ii) this is concave on the convex hull of labellings, and its linearisation at χ is $-\frac{2}{\sqrt{\tau}} \sum_k \langle \chi'_k, M y_k \rangle$. So if the assignment step *exactly* maximised $\sum_k \langle \chi'_k, M y_k \rangle$ over labellings with equal discrete area, Proposition 4.1 would give $E_\tau^M(\chi') \leq E_\tau^M(\chi)$ at every step, constraint included — the discrete form of the auction-dynamics theorem of [17].

7.2 Why the implementation is outside it

Two hypotheses fail, independently.

1. **The pairing.** The assignment maximises $\sum_k \langle \chi'_k, D y_k \rangle = \sum_k \chi_k'^\top D A_\tau \chi_k$ with the *lumped* D , while y_k was produced with the *consistent* M , and $D A_\tau = D(M + \tau K)^{-1} M$ is not symmetric. Write $D A_\tau = S + Z$ with S symmetric and Z skew; $\|Z\|_F / \|D A_\tau\|_F$ is 2×10^{-3} on the 40×36 and 60×52 torus meshes and $0.9\text{--}1.2 \times 10^{-3}$ on the 100×96 production base. The lumped energy $E_\tau^D = \tau^{-1/2} [|S| - \sum_k \chi_k^\top S \chi_k]$ (the skew part drops out of a quadratic form) is concave iff $S \succeq 0$, which is not guaranteed but was true on every mesh tested at the production τ : smallest eigenvalue $+4 \times 10^{-6}$ (40×36 , $N = 10$; 60×52 , $N = 100$) and $+6 \times 10^{-7}$, $+2 \times 10^{-6}$ (100×96 at $N = 100, 400$), against largest eigenvalues of 3×10^{-3} to 2×10^{-2} . The operative failure is elsewhere: the linearisation of $-\chi^\top S \chi$ at χ is $-2S\chi$, whereas the assignment maximises $\langle \chi', (S + Z)\chi \rangle$. The step is therefore the

exact minimise-the-linearisation step of Proposition 4.1 for E_τ^D perturbed by the skew term $\langle \chi', Z\chi \rangle$. The matrix-norm figure above is an upper bound on that perturbation, not its size: at an actual pair of labellings (the geodesic balanced init at the project seed, then one full-budget balanced step) $\langle \chi', Z\chi \rangle / \langle \chi', S\chi \rangle$ is 1.1×10^{-5} on 40×36 and 2.0×10^{-5} on 60×52 , and the skew term changes the unconstrained argmax at one vertex of 1,440 and one of 3,120 respectively (numerics.yaml: pairing_cases[*].labelling_pair). All eigenvalue and asymmetry figures in this item: pairing_cases[*]. Monotonicity of E_τ^D can fail only through that term and through inexactness (next item); tracking E_τ^M instead is simply the wrong pairing, since its gradient $-2MA_\tau\chi$ is not what the assignment sees at all.

2. **Exactness.** [17] obtain a theorem because the auction solves the assignment exactly (to ε -complementary slackness). solve_dual_offsets returns the best iterate of a subgradient method with a residual area error, and the early-stopped in-loop solves are further from optimal by design.

Report 08 records the consequence: E_τ^D increases on 4 of 96 active steps at $N = 300$, consistent with two small perturbations of an otherwise monotone scheme. Neither energy is therefore a gate. The discrete forms of the dissipation statement exist in the literature and are the right reference here: [12, §4.4] state that two-phase graph MBO with a symmetric weight matrix W decreases its energy at every step when W is positive semi-definite, and exhibit a $W = A + \lambda I$ ($\lambda < 2$) for which it cycles between two configurations (their Example 3, Fig. 3); their numbered Propositions 4.7–4.9 concern modified algorithms with split weights and are not the statement used here. For the constrained multiclass scheme [17, §5.2] state the same condition (“as long as W is positive semi-definite, the graph heat content is concave”), which is Proposition 4.1 applied to a graph energy.

7.3 The inequality that survives

Proposition 7.1 (Descent inequality). *Let ω' be any labelling of the form (18) for some $\psi \in \mathbb{R}^N$ — exact or not — and ω the previous labelling. Then*

$$\sum_k \langle \chi'_k, Dy_k \rangle - \sum_k \langle \chi_k, Dy_k \rangle \geq \sum_k \psi_k (T_k - T'_k), \quad (21)$$

where T_k, T'_k are the old and new discrete areas.

Proof. For every vertex i , the argmax gives $y_{i\omega'(i)} + \psi_{\omega'(i)} \geq y_{i\omega(i)} + \psi_{\omega(i)}$. Multiply by $v_i > 0$ and sum: the y terms give the left side of (21), and $\sum_i v_i (\psi_{\omega(i)} - \psi_{\omega'(i)}) = \sum_k \psi_k (T_k - T'_k)$. $\square$

The right-hand side is exactly the price of inexactness: at perfect balance $T_k = T'_k = \bar{A}$ it vanishes and (21) becomes the textbook monotone step for the lumped linearisation; otherwise it scales like $\|\psi\| \times (\text{area error})$: the per-level maxima recorded in report 08’s data.yaml run from 1.3×10^{-4} ($N = 100$) to 2.8×10^{-2} ($N = 1000$ on $V = 114,144$, level 1), relative to $|v^\top y_\omega|$. Because (21) needs neither concavity nor exactness, a violation beyond floating point can only mean the returned labels are not the argmax at the returned ψ , or a unit mismatch between them; that makes it worth gating on, and the measured violation is $\leq 6.1 \times 10^{-8}$.

Code: energies and the gate

```
src/partition/mbo_auction.py:      lyapunov_energy(y, labels, v, total_area, tau,
inner_consistent) returns E_lumped =  $E_\tau^D$  and E_consistent =  $E_\tau^M$ ; descent_slack(y, v,
psi, labels_old, labels_new, n_cells) returns lhs, rhs of (21) and violation_rel =
(rhs-lhs)/scale. testing/test_mbo_auction.py gate G4 asserts (21).
```

7.4 A pairing that would restore the theorem (not implemented)

Remark 7.1 (Lumped–lumped pairing). Replace (9) by the mass-lumped step $(D + \tau K) y_k = D\chi_k$, i.e. $A_\tau^D = (D + \tau K)^{-1}D$, and keep the assignment as it is. Then $DA_\tau^D = D(D + \tau K)^{-1}D$ is symmetric positive definite, the lumped energy E_τ^D is concave, its linearisation is the objective the assignment already maximises, and with an exact assignment Proposition 4.1 applies verbatim. Mass lumping is a standard, first-order-consistent variant of the P1 heat step [9]; Propositions 5.1–5.4 hold unchanged with $L = D^{-1}K$. This changes the diffusion operator and therefore every measured number, so it is a separate experiment with its own falsifiers; it is recorded here because it is the one-line change that removes the skew term of §7.2(1) exactly — with $W = D(D + \tau K)^{-1}D$, which is symmetric positive definite, the lumped energy is the graph heat content of [17, §5.2] and the assignment objective is its linearisation, so Proposition 4.1 applies — leaving only the inexactness of §7.2(2). Since the skew term is $\lesssim 2 \times 10^{-3}$ in matrix norm and $\sim 10^{-5}$ at an actual labelling pair, the experiment would be isolating a small effect.

8 The time step: a two-sided window

8.1 The rule

Per mesh level,

$$\tau = \min((c h_{\text{mean}})^2, (\rho R_{\text{cell}})^2), \quad c = 4, \quad \rho = 1, \quad (22)$$

so $\sqrt{\tau}$ is c mean edges unless that would exceed the cell radius. The two branches are the two sides of the window; the constants are pre-registered and the measured band that passes every gate is in report 08 and CLAUDE.md.

8.2 Freeze

A vertex’s label can change only if the diffused field crosses its neighbour’s at that vertex, which requires the continuum interface to move at least a mesh spacing. With Proposition 5.4 and $\kappa \sim 1/R_{\text{cell}}$ for a cell-scale feature,

$$\delta = \frac{\tau}{2R_{\text{cell}}} \gtrsim h \iff \tau \gtrsim 2h R_{\text{cell}} \iff c \gtrsim \sqrt{2R_{\text{cell}}/h} \quad \text{on the uncapped branch.} \quad (23)$$

The implementation reports the same criterion without the factor 2 (it was written with the Gaussian displacement $\tau\kappa$): `c_lo_hmean` = $\sqrt{R_{\text{cell}}/h_{\text{mean}}}$ and `c_lo_hmax_local` = $\sqrt{R_{\text{cell}}/h_{\text{max}}}$. The factor shifts the threshold by $\sqrt{2}$, inside the margin the constants were chosen with (the frozen $c = 2$ / converged $c = 4$ pair is the NC3-CAL control of `testing/test_mbo_auction.py`; report 08 §3 cites [25] for the freeze regime); it is noted, not corrected, because the ratio is a diagnostic and every measured run quotes the reported value. Both are order-of-magnitude statements: the expansion behind Proposition 5.4 assumes $h \ll \sqrt{\tau} \ll R_{\text{cell}}$, and at the coarsest levels neither inequality is strong. The rigorous statement in this direction is for graphs: [25, Theorem 4.2] proves the two-phase graph scheme is stationary for every initial set when $\tau < \log(3/2)/\rho(\Delta) \approx 0.4/\rho(\Delta)$, ρ the spectral radius. For a mesh Laplacian $\rho(L) \sim C/h^2$, so that bound is $\sqrt{\tau} \lesssim 0.6h/\sqrt{C}$ — a much weaker (necessary-for-motion) condition than (23), which involves R_{cell} .

8.3 Over-merge

On the other side, when $\sqrt{\tau}$ approaches R_{cell} a cell’s own diffused indicator no longer dominates at its own centre. From the mixture, for a disc of radius R the diffused indicator at the centre is

$$y(0) = \int_0^\infty \frac{e^{-t/\tau}}{\tau} (1 - e^{-R^2/4t}) dt = 1 - x K_1(x), \quad x = \frac{R}{\sqrt{\tau}}, \quad (24)$$

(K_1 the modified Bessel function; the Gaussian value is $1 - e^{-x^2/4}$). At the cap $\rho = 1$, $x = 1$ and $y(0) = 1 - K_1(1) = 0.398$: the cell’s own signal at its core is below $\frac{1}{2}$, i.e. the *sum* of its neighbours’ diffused indicators already exceeds it there. The argmax compares against the *largest* neighbour, so this does not by itself imply core loss; it marks the scale at which core loss becomes possible. At $x = 2$ and 4 , $y(0) = 0.720$ and 0.950 . This is why $\rho = 1$ is called the *loosest defensible* cap: it is the point at which the diffusion has spread over the whole cell, and anything tighter is a preference. Assignment quality cannot see over-merging — the balanced threshold hits its area target by construction whatever the geometry — so the over-merge instruments are geometric (§11).

8.4 The reported ratios, and anisotropy

`tau_diagnostics` reports $\sqrt{\tau}/h_{\text{mean}}$, $\sqrt{\tau}/h_{\text{max}}$, $\sqrt{\tau}/R_{\text{cell}}$, the two freeze constants, `cap_active`, and the *freeze ratio*

$$\frac{\sqrt{\tau}/h_{\text{max}}}{\sqrt{R_{\text{cell}}/h_{\text{max}}}} = \sqrt{\frac{\tau}{R_{\text{cell}} h_{\text{max}}}} = \begin{cases} c h_{\text{mean}}/\sqrt{R_{\text{cell}} h_{\text{max}}} & \text{uncapped,} \\ \sqrt{R_{\text{cell}}/h_{\text{max}}} & \text{capped,} \end{cases} \quad (25)$$

which is the quantity `CLAUDE.md` calls the τ freeze ratio. Two consequences follow from (25) and are borne out by the runs. On the capped branch the ratio *falls* as cells shrink relative to the mesh ($R_{\text{cell}} \propto N^{-1/2}$), which is why the standard 100×96 base level drops below 1 at $N = 750$ and 1000; on the uncapped branch it falls as cells *grow* relative to the mesh, which is why a well-resolved low- N configuration runs at 0.5–0.6 and passes every gate — the ratio is not a defect indicator on its own, and the measured band is the reference. The mesh is anisotropic ($h_{\text{max}}/h_{\text{mean}} = 1.81$ on the torus ladder), so a mean-based figure flatters the coarse band: the rule is on h_{mean} and the freeze ratio is on h_{max} , deliberately.

8.5 The continuation probe

E_τ depends on the labels alone once τ is fixed, so a level that froze and a level that converged both show a flat energy tail. The probe `tau_continuation_probe` re-runs one balanced threshold with $\tau' = f\tau$, $f \in \{2, 4\}$, and evaluates the resulting labels at the *level’s* τ : if a wider blur reaches a labelling with materially lower E_τ and moves a material fraction of vertices, the level was pinned. Because a wider blur always re-tiebreaks some boundary vertices and shaves some energy, the verdict needs calibrated thresholds (`nc3_K`, `nc3_f_min`), and until they are pinned by the NC3-CAL control the probe records and reports `pinned = None` rather than a verdict. A pinned level would be annealed ($\tau \leftarrow 1.5\tau$, at most three times) and re-run. With the shipped defaults `nc3_K = nc3_f_min = None` the verdict is always `None`, so the anneal branch is unreachable and every measured run reports `n_anneals = 0` vacuously — the probe records, it does not yet decide.

Code: the window

```
src/partition/mbo_auction.py: tau_diagnostics(mesh, n_cells, cfg, tau_override) implements (22) and reports every ratio of §8.4; tau_continuation_probe is §8.5; MBOConfig.tau_c, rho, anneal_factor, max_anneals, probe_factors, nc3_K, nc3_f_min. Only tau_c, rho and max_iters are config-reachable (mbo: block); the rest define the measurement protocol.
```

9 Initialisation

The first labelling is one balanced assignment of geodesic scores:

1. N sites by greedy farthest-point sampling in the *ambient* Euclidean metric of $\mathbb{R}^3$ (first site from the seeded RNG, each next site the vertex farthest from the current set) — `farthest_point_sampling`, shared with the PGD seeded init so both start from the same site layout;
2. d_{ik} = shortest-path distance from site k to vertex i on the edge graph of the mesh, edge weights the Euclidean edge lengths, by N single-source Dijkstra searches returning the full $V \times N$ matrix (`dijkstra` with `indices=sites`) — a graph metric that *over*-estimates the geodesic distance (paths are confined to edges) but respects the surface topology, since it is built from `faces`; the heat method of [7] is not what is implemented;
3. scores $s_{ik} = -d_{ik}^2$ and one full-budget balanced assignment.

With $-d^2$ scores, (18) is a *power diagram* of the graph metric with weights ψ (doc 09, §8; [3]): the capacity-constrained Voronoi diagram that [6] also use as an initialisation, computed here by the transportation dual rather than by their gradient method, and on a surface. It is also, by construction, step 0 of approach C (capacity-constrained geodesic Lloyd), which is what makes the attribution control of report 08 possible: running the init alone on the finest mesh isolates what the dynamics add.

The balanced start is a **hard requirement**, measured rather than argued: from a cold unbalanced Voronoi labelling the first balanced threshold leaves a worst cell at 68.16% area deviation, from the balanced one at 1.97% (`docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md`, Phase A, and the module docstring; report 08 does not carry these two numbers).

Code: initialisation

```
src/partition/mbo_auction.py:                geodesic_bridged_init(mesh, n_cells,
seed);                edge_graph(mesh);                src/optimization/initialization.py:
farthest_point_sampling(vertices, n_seeds, seed); scipy.sparse.csgraph.dijkstra.
```

10 The mesh ladder

Levels are the same mesh ladder Phase 1 uses. Within a level, steps repeat until the fraction of vertices that change label is below `churn_tol` = 10^{-4} for `patience` = 3 consecutive steps, or `max_iters` = 200 is hit (never, in the measured runs: the most any level took is 66). Between levels the labelling is carried by *nearest embedded vertex*: each fine vertex takes the label of the nearest coarse vertex in $\mathbb{R}^3$ (`cKDTree`). Labels are interpolated rather than densities, because a hard partition has no density to interpolate and the nearest-vertex map of a labelling is itself a labelling. What the transfer preserves: each cell's territory up to a band of width $O(h_{\text{coarse}})$ along its boundary, hence its area up to $O(h_{\text{coarse}} \cdot \text{Per}(\Omega_k))$ — the first balanced threshold on the fine level removes that error. What it does not preserve: connectivity is not guaranteed across the transfer (a thin neck can be cut), and τ is recomputed from (22) on the new mesh, so the effective diffusion length shrinks with h on the uncapped branch. Report 08 records fragmented cells born at one level and healed at the next; the ladder is what heals them.

Code: ladder

```
src/partition/mbo_auction.py:                run_mbo_ladder(meshes, n_cells, cfg, init_only),
mbo_level(mesh, labels, n_cells, cfg, level, run_probe),                _run_steps,
interpolate_labels(old_vertices, new_vertices, labels).
```

11 Instruments

None of these is a gate on the dynamics; they are what the reports quote.

- **Energies.** E_τ^D (`E_lumped`, primary because its linearisation differs from the assignment objective only by the skew term of §7.2) and E_τ^M (`E_consistent`), both by (20) with the respective weight, and both $\rightarrow \frac{1}{2} \sum_k \text{Per}(\Omega_k)$ by Proposition 5.3. Descriptive trends only (§7.2).
- **Label boundary length.** The midpoint-cut construction: a triangle with two labels contributes the segment joining the midpoints of its two bi-labelled edges, one with three labels contributes three segments from those midpoints to the centroid (`label_boundary_length`, `src/partition/balanced_readout.py`). It counts each interface once; the Phase 2 perimeter counts each twice, and the measured ratio between them on an unoptimised labelling is 1.999 (`CLAUDE.md`).
- **Compactness.** $Q_k = P_k^2 / (4\pi A_k)$ with P_k the per-cell boundary length (same construction, each segment credited to both cells) and $A_k = T_k$; $Q = 1$ is a disc. Over-merging raises Q .
- **Core loss.** The number of cells whose own diffused-score peak vertex, $\arg \max_i y_{ik}$, is not in their territory: the most direct over-merge symptom, since it says the cell's signal was swamped before the threshold was taken.
- **The three Phase 1 gates,** run on every level's labels through the same harness path as the final labels (`run_gates`). For one-hot labels the dormant gate is vacuous and the area gate is the solver check of §6.3; only connectivity has content, because nothing in §6 enforces it.

Code: instruments

```
src/partition/mbo_auction.py:      lyapunov_energy,      per_cell_boundary_lengths,
compactness, core_loss; src/partition/balanced_readout.py:  label_boundary_length;
src/partition/arm_harness.py: run_gates, labels_to_one_hot.
```

Summary table

Quantity	Status	Implementation	
Diffusion step $(M + \tau K)y_k = M\chi_k$	derived (backward Euler of the P1 heat equation)	<code>make_diffusion_solver</code> , <code>diffuse_indicators</code>	
Mixture representation (10)	proved	— (analysis)	
$\sum_k y_k = \mathbf{1}$; MA_τ SPD; no discrete positivity	proved	—	
Continuum kernel $\frac{1}{2\pi\tau}K_0(r/\sqrt{\tau})$, marginal (12)	derived	—	
Interface constant $c_{K_\tau^{\text{res}}} = \frac{1}{2}$ (Gaussian $1/\sqrt{\pi}$)	proved; Γ -limit cited [12] (flat)	<code>lyapunov_energy</code> $\frac{1}{2}\sum_k \text{Per}(\Omega_k)$	approximates
Per-step displacement $\tau\kappa/2$	proved (leading order); exact disc solution (16); mesh at 0.91–0.97 of it and 0.48–0.54 of the Gaussian, placement-averaged	§5.6	
Balanced threshold $\arg \max_k (y_{ik} + \psi_k)$	derived (transportation dual, doc 09); $N=2$ case is Ruuth–Wetton	<code>balanced_assign</code> <code>solve_dual_offsets</code>	→
Dissipation with exact assignment	theorem (cited); the implemented step is the exact MM step for E_τ^D up to a $\lesssim 10^{-3}$ skew term and the inexact dual	—	
Descent inequality (21)	proved for any argmax labelling	<code>descent_slack</code> ; gate G4	
Lumped–lumped pairing restores the theorem	stated; not implemented	—	
τ rule (22); freeze (23); over-merge (24)	derived (order of magnitude; $y(0) = 1 - xK_1(x)$ exact for the continuum kernel)	<code>tau_diagnostics</code>	
Freeze ratio (25) and its two regimes	derived	<code>tau_diagnostics</code>	
Continuation probe	protocol; verdict uncanceled until NC3-CAL	<code>tau_continuation_probe</code>	
Init as power diagram of the graph metric	derived	<code>geodesic_balanced_init</code>	
Ladder transfer error $O(h \text{Per})$	stated	<code>interpolate_labels</code>	
E_τ , boundary length, Q_k , core loss	instruments	§11	

Related documents

- Measurements, pre-registration and the negative controls:
docs/experiments/08-mbo-auction-dynamics/ and docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md (Phase A).
- The transportation dual and the readout it was built for:
docs/math/09-balanced-readout/.
- The P1 discretisation and the Phase 1 energy this replaces:
docs/math/06-phase1-energy-discretization/.
- Taxonomy of the high- N approaches:
docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md.

- Held copies of every cited paper: docs/papers/README.md.

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